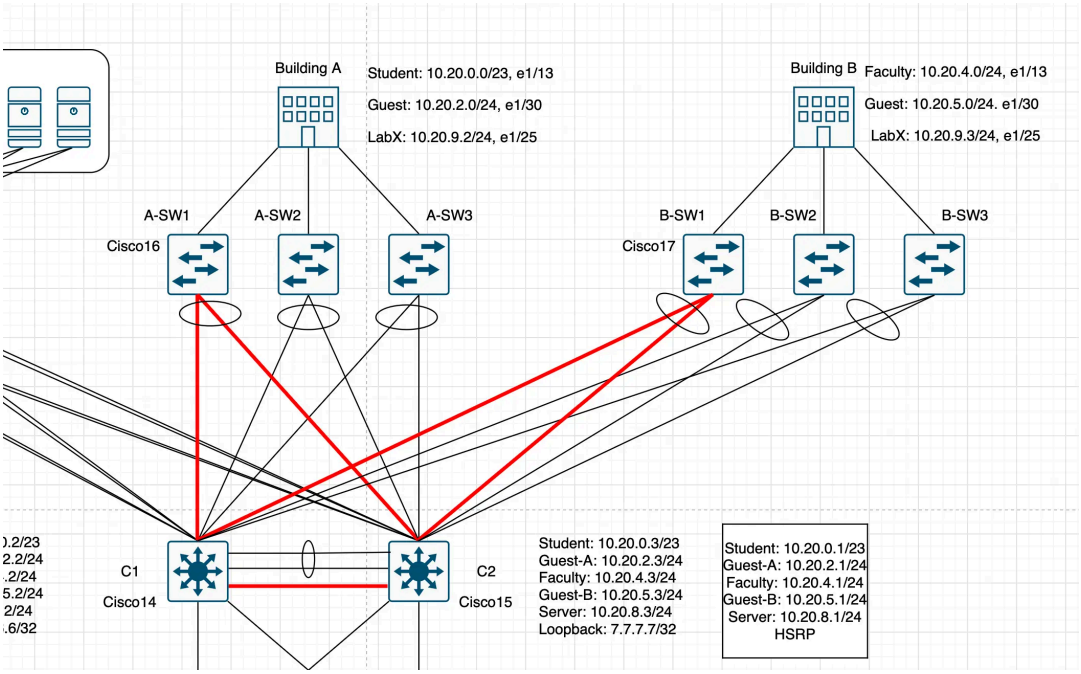
VXLAN Change Request Report



New VLAN for LabX: 100

Host at Building A: 10.20.9.2

Host at Building B: 10.20.9.3

Our previous configuration

Here we configured A-SW1 -> C1, A-SW1 -> C2, B-SW1 -> C1, B-SW1 -> C2 as trunks for VLAN 100.

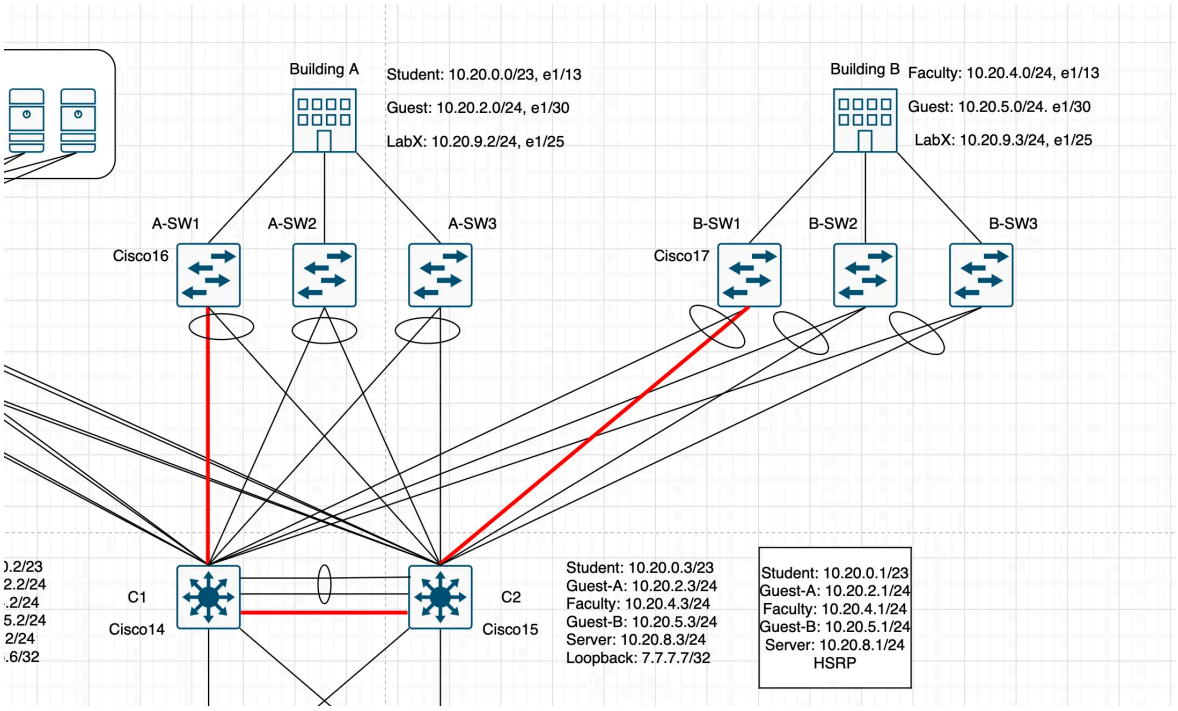
VTEPs: C1, C2

C1 -> C2 link is layer 3.

But the problem was A-SW1 sends packet to C1. C1 already has a trunk link to B-SW1. So it can send to B-SW1. And so it can update its MAC table with that value. Also, C1 has VTEP configured for VLAN 100. So that flow also runs from C1 -> C2 -> B-SW1. This also updates the mac table. The fast change in mac addresses flag a loop and error-disables the port on C1 connected to A-SW1.

Similar thing happens on C2 and error-disables its port connected to A-SW1. Essentially, no packet coming from A-SW1 reached C1 and C2.

Solution

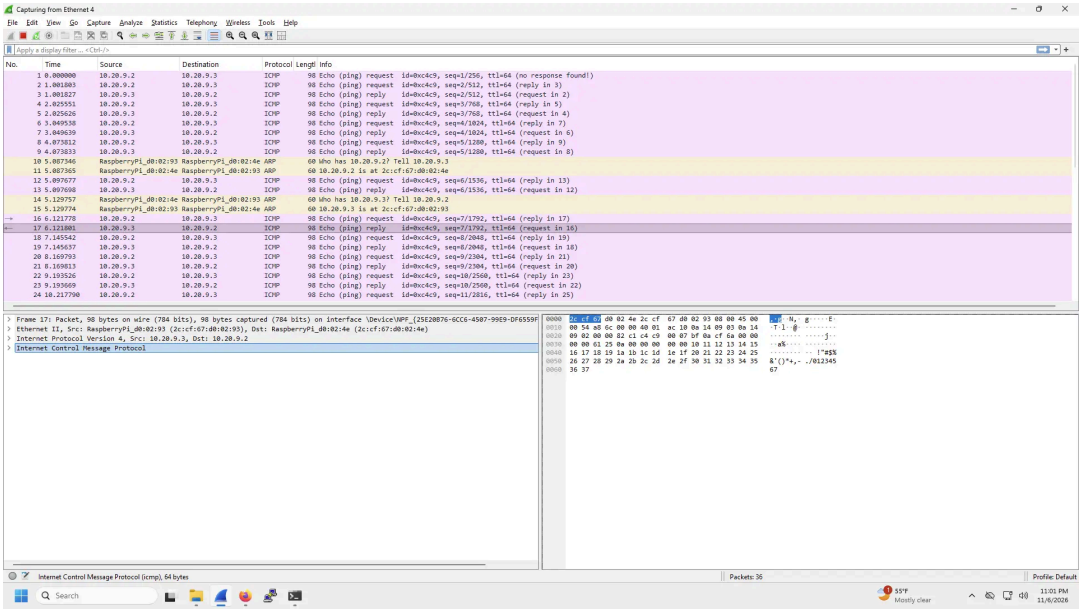


Since C2 is remote VTEP for A-SW1 and C1 is remote VTEP for B-SW1. So the links connecting them should not act as trunk links for VLAN 100.

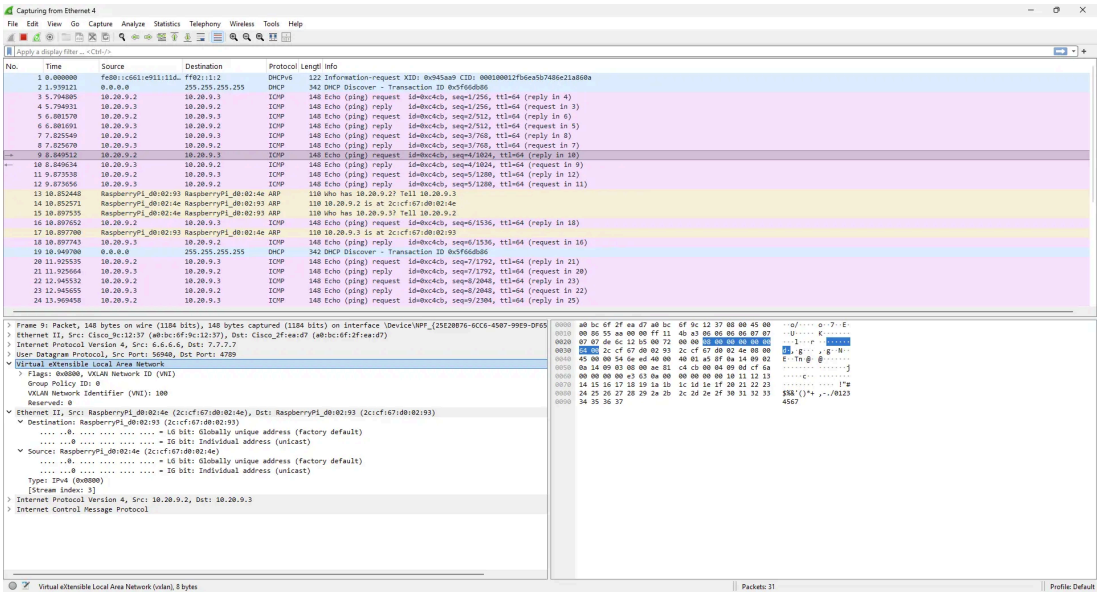
Now everything works.

Wireshark Captures

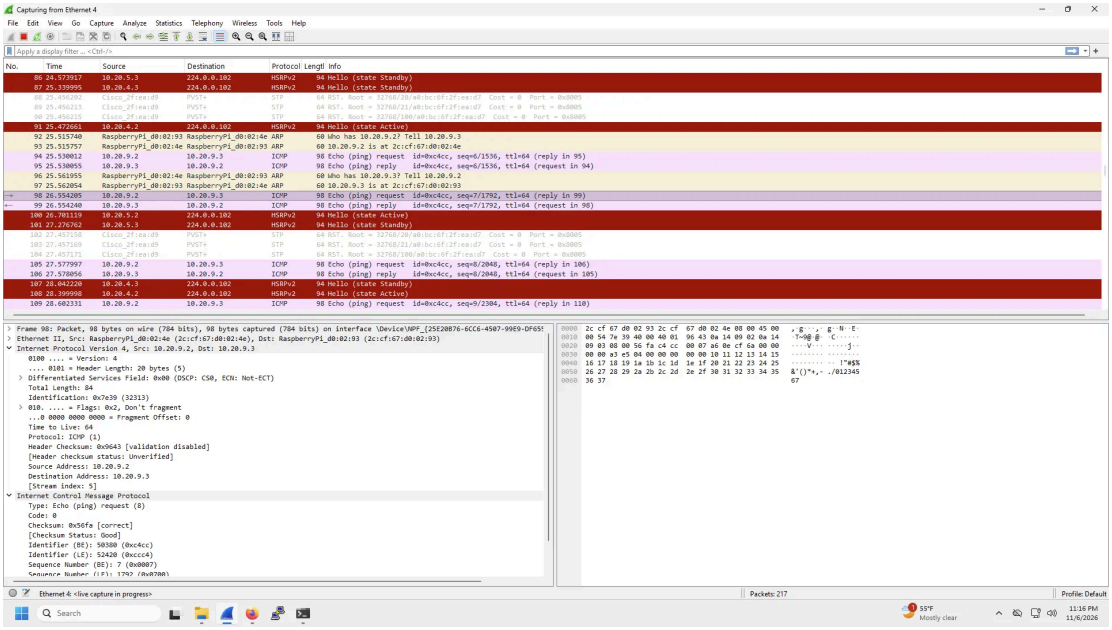
Host to A-SW1



C1 to C2



C2 to B-SW1


Capturing from Ethernet 4
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
86 24.573917 10.20.5.3 224.0.0.102 HSRPv2 94 Hello (state Standby)
87 25.339995 10.20.4.3 224.0.0.102 HSRPv2 94 Hello (state Standby)
88 25.456202 Cisco_2f:ea:d9 PVST+ STP 64 RST. Root = 32768/20/a0:bc:6f:2f:ea:d7 Cost = 0 Port = 0x8005
89 25.456213 Cisco_2f:ea:d9 PVST+ STP 64 RST. Root = 32768/21/a0:bc:6f:2f:ea:d7 Cost = 0 Port = 0x8005
90 25.456215 Cisco_2f:ea:d9 PVST+ STP 64 RST. Root = 32768/100/a0:bc:6f:2f:ea:d7 Cost = 0 Port = 0x8005
91 25.472661 10.20.4.2 224.0.0.102 HSRPv2 94 Hello (state Active)
92 25.515740 RaspberryPi_d0:02:93 RaspberryPi_d0:02:4e ARP 60 Who has 10.20.9.2? Tell 10.20.9.3
93 25.515757 RaspberryPi_d0:02:4e RaspberryPi_d0:02:93 ARP 60 10.20.9.2 is at 2c:cf:67:d0:02:4e
94 25.530012 10.20.9.2 10.20.9.3 ICMP 98 Echo (ping) request id=0xc4cc, seq=6/1536, ttl=64 (reply in 95)
95 25.530055 10.20.9.3 10.20.9.2 ICMP 98 Echo (ping) reply id=0xc4cc, seq=6/1536, ttl=64 (request in 94)
96 25.561955 RaspberryPi_d0:02:4e RaspberryPi_d0:02:93 ARP 60 Who has 10.20.9.3? Tell 10.20.9.2
97 25.562054 RaspberryPi_d0:02:93 RaspberryPi_d0:02:4e ARP 60 10.20.9.3 is at 2c:cf:67:d0:02:93
98 26.554205 10.20.9.2 10.20.9.3 ICMP 98 Echo (ping) request id=0xc4cc, seq=7/1792, ttl=64 (reply in 99)
99 26.554240 10.20.9.3 10.20.9.2 ICMP 98 Echo (ping) reply id=0xc4cc, seq=7/1792, ttl=64 (request in 98)
100 26.701119 10.20.5.2 224.0.0.102 HSRPv2 94 Hello (state Active)
101 27.276762 10.20.5.3 224.0.0.102 HSRPv2 94 Hello (state Standby)
102 27.457158 Cisco_2f:ea:d9 PVST+ STP 64 RST. Root = 32768/20/a0:bc:6f:2f:ea:d7 Cost = 0 Port = 0x8005
103 27.457169 Cisco_2f:ea:d9 PVST+ STP 64 RST. Root = 32768/21/a0:bc:6f:2f:ea:d7 Cost = 0 Port = 0x8005
104 27.457171 Cisco_2f:ea:d9 PVST+ STP 64 RST. Root = 32768/100/a0:bc:6f:2f:ea:d7 Cost = 0 Port = 0x8005
105 27.577997 10.20.9.2 10.20.9.3 ICMP 98 Echo (ping) request id=0xc4cc, seq=8/2048, ttl=64 (reply in 106)
106 27.578056 10.20.9.3 10.20.9.2 ICMP 98 Echo (ping) reply id=0xc4cc, seq=8/2048, ttl=64 (request in 105)
107 28.042220 10.20.4.3 224.0.0.102 HSRPv2 94 Hello (state Standby)
108 28.399998 10.20.4.2 224.0.0.102 HSRPv2 94 Hello (state Active)
109 28.602331 10.20.9.2 10.20.9.3 ICMP 98 Echo (ping) request id=0xc4cc, seq=9/2304, ttl=64 (reply in 110)
Frame 98: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{25E20B76-6CC6-4507-99E9-DF65
Ethernet II, Src: RaspberryPi_d0:02:4e (2c:cf:67:d0:02:4e), Dst: RaspberryPi_d0:02:93 (2c:cf:67:d0:02:93)
Internet Protocol Version 4, Src: 10.20.9.2, Dst: 10.20.9.3
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0x7e39 (32313)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0x9643 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.20.9.2
Destination Address: 10.20.9.3
[Stream index: 5]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x56fa [correct]
[Checksum Status: Good]
Identifier (BE): 50380 (0xc4cc)
Identifier (LE): 52420 (0xccc4)
Sequence Number (BE): 7 (0x0007)
0000 2c cf 67 d0 02 93 2c cf 67 d0 02 4e 08 00 45 00
0010 00 54 7e 39 40 00 40 01 96 43 0a 14 09 02 0a 14
0020 09 03 08 00 56 fa c4 cc 00 07 a6 0e cf 6a 00 00
0030 00 00 a3 e5 04 00 00 00 00 00 10 11 12 13 14 15
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35
0060 36 37
Ethernet 4: <live capture in progress>
Packets: 217
Profile: Default
Search
55°F
Mostly clear
11:16 PM
11/6/2026